COM-480 - Data Visualization

BeanMap – Milestone 2

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1 Project Overview: *The Hidden Coffee Chain*

Coffee is one of the most traded agricultural commodities in the world, yet the geography of its economic value remains largely invisible. While producing countries export raw beans, much of the value is captured later in the supply chain through roasting, processing, packaging, and re-export. Traditional trade maps therefore fail to show where transformation actually happens.
BeanMap addresses this gap by visualizing the global coffee value chain as a transformation process rather than a simple exchange network. Instead of focusing only on bilateral trade links, the visualization highlights countries that import raw coffee and export processed coffee, revealing where value is created along the chain.

The central metric driving the visualization is the Transformation Index, defined as the ratio between processed-coffee export value and total coffee export value for each country and year. This indicator distinguishes raw-bean producers from processing hubs and downstream consumer markets, and allows users to observe how the geography of transformation has evolved from the 1980s to today.

Our objective is to make the hidden structure of the global coffee supply chain visible through an interactive narrative combining geography, flows, and temporal evolution.

2 Target Audience and Storytelling Goals

BeanMap targets a general audience interested in global trade, sustainability, and economic geography, as well as students seeking to better understand how value chains operate beyond production statistics. Coffee provides an accessible case study while illustrating structural inequalities between producing and processing regions. The visualization supports two complementary modes of exploration:

- a global overview showing where coffee transformation occurs;
- country-level detail explaining how individual countries participate in the supply chain.

Rather than presenting isolated charts, the interface encourages users to move between spatial, temporal, and network perspectives of trade. The goal is that users leave the visualization with a clear answer to the guiding question: Who really transforms the coffee bean?

3 Data Source and Core Metric

The visualization relies on the FAO Detailed Trade Matrix (1986–2024), which provides bilateral trade statistics between reporting countries and partners for agricultural commodities, including both quantities and trade values. We restrict the dataset to two coffee product categories:

- *Coffee, green*
- *Coffee, roasted (including decaffeinated)*

These categories represent raw beans and processed coffee products, enabling interpretation of trade flows as stages of transformation within the value chain. From these variables we construct the Transformation Index:

$$\text{Transformation Index} = \frac{\text{Processed coffee export value}}{\text{Total coffee export value}}$$

This index measures whether a country primarily exports raw coffee or higher-value processed products. Low values indicate producing countries, while higher values identify transformation hubs that capture additional value before re-exporting. Using this metric across the full time range allows us to track how the geography of coffee processing has shifted over four decades.

4 Visualization Concept and Layout Design

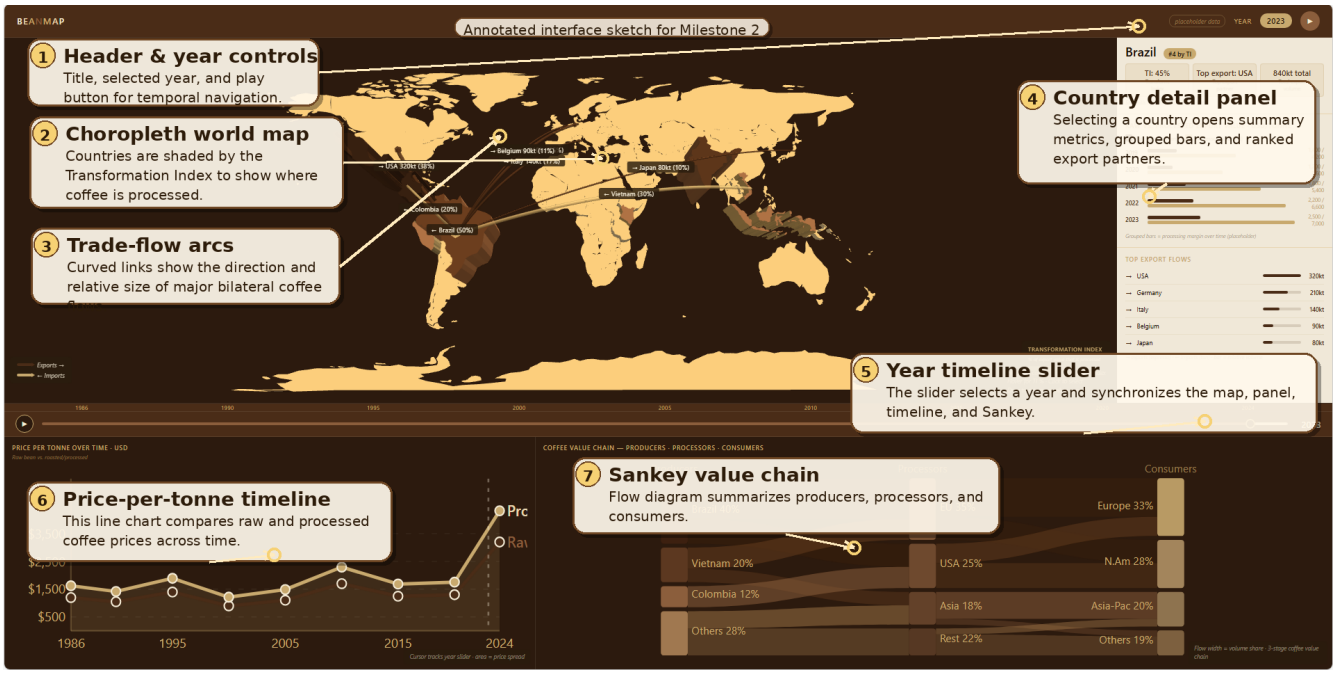

Figure 1: Annotated layout sketch of the BeanMap interface showing the coordinated visualization components: choropleth map, trade-flow arcs, country detail panel, year slider, price timeline, and Sankey value-chain diagram.

BeanMap is implemented as a single-page interactive visualization organized around coordinated views supporting exploration at multiple scales. As shown in Figure 1, the visualization is structured around six coordinated components: (1) a choropleth world map displaying the Transformation Index, (2) animated bilateral trade-flow arcs, (3) a country detail panel comparing raw and processed exports, (4) a Sankey diagram summarizing producer–processor–consumer roles, (5) a global price timeline showing long-term evolution of raw and processed coffee values, and (6) a year slider enabling synchronized temporal navigation across all views.

5 Interaction Design

Interaction enables users to explore the coffee value chain across spatial and temporal dimensions simultaneously. The year slider acts as the main coordination mechanism linking all components. Changing the selected year updates the choropleth shading, trade flows, Sankey diagram, and country detail panel consistently across views. Users can also select countries directly on the map, dynamically updating

export comparisons and processing indicators in the detail panel. Animated trade flows further support temporal exploration by making changes in exchange patterns immediately visible.

Together, these interactions support exploratory analysis while preserving a clear narrative structure.

6 Technology Stack and Implementation Strategy

The visualization is implemented using D3.js (version 7) together with TopoJSON geographic boundaries, allowing the application to run within a single HTML page without external frameworks.

Geographic projections are handled using `d3.geoNaturalEarth1`, while choropleth coloring relies on sequential scales from `d3.scaleSequential`. Trade-flow arcs are generated from interpolated geographic paths between country centroids, and the Sankey diagram is implemented using the `d3-sankey` extension.

All components share a common application state containing the selected year and selected country. Updates to this state trigger coordinated re-rendering through a D3 event dispatch mechanism. Preprocessed CSV files generated during exploratory analysis are loaded at runtime using `d3.csv`, enabling efficient filtering by country and year.

7 Implementation Roadmap

7.1 Minimal Viable Product

The MVP includes the choropleth world map, year slider, country detail panel, animated bilateral trade-flow arcs, Sankey diagram, global timeline, and play/pause animation. Together, these elements provide a complete interactive narrative explaining where coffee transformation occurs and how it evolves over time.

7.2 Planned Enhancements

Optional extensions intended to improve usability include:

- hover tooltips displaying country names and Transformation Index values,
- filters for isolating coffee product categories,
- comparison mode showing two years side by side,
- short introductory scrollytelling guidance,
- adjustable node-selection thresholds in the Sankey diagram.

8 Prototype Status

A functional prototype of BeanMap is already available online at https://beanmap.vercel.app/ and implements the core structure of the visualization. The current version includes the choropleth world map, temporal navigation through the year slider, animated trade flows, and coordinated updates across views.

The prototype demonstrates the feasibility of synchronizing multiple components through a shared application state and confirms that the Transformation Index effectively reveals global processing patterns.

Future work will focus on refining interaction details, improving performance and visual clarity, and completing the country detail panel and Sankey diagram in preparation for Milestone 3.

9 Methodological Alignment with Course Material

Visualization component	**Tools / Libraries**	**Relevant lectures**
Choropleth world map	D3.js, TopoJSON	Maps, Maps in Practice, D3 introduction
Trade-flow arcs	D3 geo paths, SVG transitions	Maps, Interaction, Animation
Country detail panel (bar chart)	D3 scales and SVG rectangles	Marks & Channels, Perception, Interaction
Global price timeline	D3 line generator and axes	D3, Time series visualization
Sankey diagram	d3-sankey plugin	Network visualization
Year slider and play/pause control	HTML range input, JavaScript, D3 dispatch	Interaction lectures
Data preprocessing pipeline	JavaScript, CSV, GeoJSON formats	JavaScript, Tabular Data
Narrative structure and layout	Coordinated views and storytelling structure	Storytelling lecture

Table 1: Tools and course concepts supporting the implementation of each visualization component.